

Solving Trigonometric Equations on a Restricted Interval

Answer Key

Precalculus

Quick Answers

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|--|---|
| 1. 120, 240 degrees
2. 45, 135 degrees
3. 0, 30, 150, 180 degrees
4. 30, 60, 210, 240 degrees
5. (a) factored form =
$(1 - \cos(\theta))(2 \cos(\theta) - 1)$, (b) solutions = 0,
60, 300 degrees
6. (a) the times = 1, 5 hours, — | 7. 20, 100, 140 degrees
8. (a) factored form =
$(1 - 2 \sin(\theta))(\sin(\theta) + 1)$, (b) solutions = 30,
150, 270 degrees
9. 15, 75 degrees
10. (a) all solutions = 60, 120, 240, 300 degrees,
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What is verified

12 of 14 answers machine-checked against SymPy, across 10 problems.

— marks an answer only you can judge — problems 6, 10. The worked solution states what a correct response must contain.

Curriculum

Level: Precalculus

HSF-TF.B.5 – *problems 1, 2, 3, 4, 6, 7, 9*

HSF-TF.C.8 – *problems 5, 8, 10*

Difficulty 1–5 of 5 across 14 tagged checks.

- 1.** Solve $2 \cos \theta + 1 = 0$ for $0^\circ \leq \theta < 360^\circ$.

Solution: isolate the cosine, then use the reference angle and the two quadrants where cosine is negative.

$$\cos \theta = -\frac{1}{2}, \quad \text{reference angle } 60^\circ, \quad \text{quadrants II and III.}$$

So $\theta = 180^\circ - 60^\circ = 120^\circ$ and $\theta = 180^\circ + 60^\circ = 240^\circ$.

$\theta = 120^\circ, 240^\circ$

- 2.** Solve $\sqrt{2} \sin \theta - 1 = 0$ for $0^\circ \leq \theta < 360^\circ$.

Solution: $\sin \theta = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}$, whose reference angle is 45° . Sine is positive in quadrants I and II, so $\theta = 45^\circ$ and $\theta = 180^\circ - 45^\circ = 135^\circ$.

$\theta = 45^\circ, 135^\circ$

3. Solve $\sin \theta (2 \sin \theta - 1) = 0$ for $0^\circ \leq \theta < 360^\circ$.

Solution: the product is zero exactly when one factor is zero, so treat two equations.

$$\begin{aligned} \sin \theta = 0 &\implies \theta = 0^\circ, 180^\circ \\ 2 \sin \theta - 1 = 0 &\implies \sin \theta = \frac{1}{2} \implies \theta = 30^\circ, 150^\circ \end{aligned}$$

Note that $\theta = 360^\circ$ is excluded by the half-open interval.

$\theta = 0^\circ, 30^\circ, 150^\circ, 180^\circ$

4. Cam: solve $2 \sin 2\theta - \sqrt{3} = 0$ for $0^\circ \leq \theta < 360^\circ$.

Solution: let $u = 2\theta$. As θ runs over $[0^\circ, 360^\circ)$, u runs over $[0^\circ, 720^\circ)$ — widen the interval *before* solving, or half the answers are lost.

$$\sin u = \frac{\sqrt{3}}{2} \implies u = 60^\circ, 120^\circ, 420^\circ, 480^\circ.$$

Dividing each by 2: $\theta = 30^\circ, 60^\circ, 210^\circ, 240^\circ$. All four lie in the required interval.

$\theta = 30^\circ, 60^\circ, 210^\circ, 240^\circ$

5. $2 \sin^2 \theta + 3 \cos \theta - 3 = 0$ on $0^\circ \leq \theta < 360^\circ$. (a) Factor. (b) Solve.

(a) The equation mixes $\sin^2$ with $\cos$, so replace $\sin^2 \theta$ by $1 - \cos^2 \theta$ to leave one function:

$$\begin{aligned} 2(1 - \cos^2 \theta) + 3 \cos \theta - 3 &= -2 \cos^2 \theta + 3 \cos \theta - 1 \\ &= -(2 \cos^2 \theta - 3 \cos \theta + 1) = (1 - \cos \theta)(2 \cos \theta - 1). \end{aligned}$$

$(1 - \cos \theta)(2 \cos \theta - 1)$

(b) Set each factor to zero: $\cos \theta = 1$ gives $\theta = 0^\circ$; $\cos \theta = \frac{1}{2}$ gives $\theta = 60^\circ$ and $\theta = 300^\circ$.

$\theta = 0^\circ, 60^\circ, 300^\circ$

6. Buoy: $h(t) = 6 \sin\left(\frac{\pi t}{6}\right)$ metres. (a) Times with $h = 3$ on $0 \leq t < 12$. (b) Interpret them.

(a) $6 \sin\left(\frac{\pi t}{6}\right) = 3$, so $\sin\left(\frac{\pi t}{6}\right) = \frac{1}{2}$. As t runs over $[0, 12)$ the inside angle runs over $[0, 2\pi)$ exactly once, and sine equals $\frac{1}{2}$ there at $\frac{\pi}{6}$ and $\frac{5\pi}{6}$:

$$\frac{\pi t}{6} = \frac{\pi}{6} \implies t = 1, \quad \frac{\pi t}{6} = \frac{5\pi}{6} \implies t = 5.$$

$t = 1, 5 \text{ hours}$

(b) *Instructor-judged.* A correct response interprets rather than restates: the buoy passes 3 metres twice per cycle, once rising (at $t = 1$) and once falling (at $t = 5$), because the height

peaks at 6 metres at $t = 3$ and returns. Credit an answer that says which crossing is rising and which is falling; repeating the two numbers is not enough.

7. Solve $2 \cos 3\theta - 1 = 0$ for $0^\circ \leq \theta < 180^\circ$.

Solution: let $u = 3\theta$, so u runs over $[0^\circ, 540^\circ)$.

$$\cos u = \frac{1}{2} \implies u = 60^\circ, 300^\circ, 420^\circ.$$

(The next value, 660° , is past 540° .) Dividing by 3 gives $\theta = 20^\circ, 100^\circ, 140^\circ$.

$\theta = 20^\circ, 100^\circ, 140^\circ$

8. $2 \cos^2 \theta - \sin \theta - 1 = 0$ on $0^\circ \leq \theta < 360^\circ$. (a) Factor. (b) Solve.

(a) Replace $\cos^2 \theta$ by $1 - \sin^2 \theta$ so only sine remains:

$$\begin{aligned} 2(1 - \sin^2 \theta) - \sin \theta - 1 &= -2\sin^2 \theta - \sin \theta + 1 \\ &= (1 - 2\sin \theta)(\sin \theta + 1). \end{aligned}$$

$(1 - 2\sin \theta)(\sin \theta + 1)$

(b) $1 - 2\sin \theta = 0$ gives $\sin \theta = \frac{1}{2}$, so $\theta = 30^\circ$ and 150° ; $\sin \theta + 1 = 0$ gives $\sin \theta = -1$, so $\theta = 270^\circ$ — a single solution, since -1 is the minimum of sine and is reached once per turn.

$\theta = 30^\circ, 150^\circ, 270^\circ$

9. Searchlight: solve $40 \sin 2\theta = 20$ for $0^\circ \leq \theta < 180^\circ$.

Solution: $\sin 2\theta = \frac{20}{40} = \frac{1}{2}$. With $u = 2\theta$, u runs over $[0^\circ, 360^\circ)$, where $\sin u = \frac{1}{2}$ at $u = 30^\circ$ and $u = 150^\circ$. Halving: $\theta = 15^\circ$ and $\theta = 75^\circ$.

$\theta = 15^\circ, 75^\circ$

10. $(2 \sin \theta - \sqrt{3})(2 \sin \theta + \sqrt{3}) = 0$ on $0^\circ \leq \theta < 360^\circ$. (a) Solve. (b) Explain the count.

(a) Each factor gives its own equation.

$$\begin{aligned} \sin \theta &= \frac{\sqrt{3}}{2} \implies \theta = 60^\circ, 120^\circ \\ \sin \theta &= -\frac{\sqrt{3}}{2} \implies \theta = 240^\circ, 300^\circ \end{aligned}$$

$\theta = 60^\circ, 120^\circ, 240^\circ, 300^\circ$

(b) *Instructor-judged.* A correct response explains that the product form allows *two* values of $\sin \theta$, $\frac{\sqrt{3}}{2}$ and $-\frac{\sqrt{3}}{2}$, and each value is attained in two quadrants during one full turn — two cases times two quadrants is four angles. The single linear equation permits only the positive value, so it yields two. Full credit requires that two-cases-times-two-quadrants reason; noticing only that one list is longer is not enough.